

Fixed-Dimension Approximation of Symmetric Convex Bodies by Intersections of Centered Ellipsoids

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Abstract

A question motivated by the expressiveness of generalized ellipsoids asks how well an origin-symmetric convex body in $\mathbb{R}^n$ can be approximated by an intersection of m co-centered ellipsoids. This note gives a complete answer to the fixed-dimensional asymptotic version of that question. For each fixed $n \geq 2$, let $m_n(\varepsilon)$ denote the least m such that every origin-symmetric convex body $C \subset \mathbb{R}^n$ admits positive definite matrices $Q_1, \dots, Q_m$ for which

$$K := \bigcap_{i=1}^m \{x \in \mathbb{R}^n : x^T Q_i x \leq 1\},$$

$$K \subseteq C \subseteq (1 + \varepsilon)K.$$

Then, as $\varepsilon \downarrow 0$,

$$m_n(\varepsilon) = \Theta_n(\varepsilon^{-(n-1)/2}).$$

Equivalently, for fixed n , the worst-case multiplicative error achievable with m centered ellipsoids is

$$\Theta_n(m^{-2/(n-1)}).$$

This should be read as partial progress on the full dimension-dependent problem, not as a complete high-dimensional solution: the constants hidden in $\Theta_n(\cdot)$ may depend on n . The upper bound is obtained by converting the classical fixed-dimensional Dudley–Bronshtein–Ivanov polytopal approximation theorem into intersections of positive definite centered ellipsoids. The lower bound uses a body obtained from the Euclidean ball by cutting off many shallow, separated antipodal caps. The key point is that a strict superlevel set of one quadratic form is connected in real projective space, so one ellipsoid can be responsible for at most one separated cap pair.

1 Introduction and status of the result

Ahmadi, Chaudhry, and Dibek introduced generalized ellipsoids and showed, among other things, that every intersection of m co-centered ellipsoids is representable as a generalized ellipsoid of degree at most $2m - 3$ [1]. They also use such representations, together with polytopal approximation, to approximate arbitrary symmetric convex bodies. A natural question is whether allowing intersections of co-centered ellipsoids improves the approximation tradeoff relative to the polytopal construction.

The result below answers the fixed-dimensional asymptotic version of this question. It shows that, for each fixed dimension n , intersections of centered ellipsoids have the same worst-case exponent as approximation by polytopes with facets. Thus, in fixed dimension, the continuous parameters available in each ellipsoid do not improve the worst-case exponent.

The result is not a full answer to the dimension-dependent version of the question. All constants in the estimates below are allowed to depend on n . In particular, the paper does not determine

the optimal dependence on both parameters (n, ε) , nor does it produce dimension-uniform or polynomial-in- n constants.

2 Definitions and main theorem

All convex bodies in this note are compact, convex, full-dimensional subsets of $\mathbb{R}^n$ with nonempty interior. A body C is called symmetric if $C = -C$. For a symmetric convex body C , its radial function is

$$r_C(u) := \sup\{t \geq 0 : tu \in C\}, \quad u \in \mathbb{S}^{n-1}.$$

If

$$K = \bigcap_{i=1}^m \{x \in \mathbb{R}^n : x^T Q_i x \leq 1\}, \quad Q_i \succ 0,$$

then

$$r_K(u) = \min_{1 \leq i \leq m} \frac{1}{\sqrt{u^T Q_i u}} = \frac{1}{\sqrt{\max_{1 \leq i \leq m} u^T Q_i u}}.$$

Thus the problem is equivalently one of approximating the squared reciprocal radial function $r_C(u)^{-2}$ by the maximum of finitely many positive definite quadratic forms.

For $0 < \varepsilon < 1$, define $m_n(\varepsilon)$ to be the least integer m such that every symmetric convex body $C \subset \mathbb{R}^n$ admits positive definite matrices $Q_1, \dots, Q_m$ satisfying

$$K := \bigcap_{i=1}^m \{x : x^T Q_i x \leq 1\} \subseteq C \subseteq (1 + \varepsilon)K.$$

Theorem 2.1 (Fixed-dimensional answer). *For every fixed integer $n \geq 2$ there are constants $0 < c_n < C_n < \infty$ and $\varepsilon_n > 0$ such that, for all $0 < \varepsilon < \varepsilon_n$,*

$$c_n \varepsilon^{-(n-1)/2} \leq m_n(\varepsilon) \leq C_n \varepsilon^{-(n-1)/2}.$$

For $n = 1$, one centered ellipsoid, namely one symmetric interval, gives exact approximation, so $m_1(\varepsilon) = 1$.

The theorem also gives the equivalent error-rate formulation. Define

$$\Delta_n(m) := \sup_C \inf_{Q_1, \dots, Q_m \succ 0} \inf\{\delta > 0 : K \subseteq C \subseteq (1 + \delta)K\}, \quad K = \bigcap_{i=1}^m \{x : x^T Q_i x \leq 1\},$$

where the supremum is over all symmetric convex bodies $C \subset \mathbb{R}^n$.

Corollary 2.2. *For every fixed $n \geq 2$,*

$$\Delta_n(m) = \Theta_n(m^{-2/(n-1)}) \quad (m \rightarrow \infty).$$

3 The upper bound

We use the following standard fixed-dimensional form of the classical approximation result of Dudley, Bronshtein, and Ivanov.

Lemma 3.1 (Classical outer approximation by symmetric polytopes). *For every fixed $n \geq 2$ there is a constant $A_n < \infty$ such that, for every symmetric convex body $C \subset \mathbb{R}^n$ and every $0 < \eta < 1$, there is a symmetric polytope P with at most*

$$A_n \eta^{-(n-1)/2}$$

pairs of facets such that

$$C \subseteq P \subseteq (1 + \eta)C.$$

Remark 3.2. *This is the usual fixed-dimensional Hausdorff approximation theorem, converted to a multiplicative statement by an affine normalization such as John's ellipsoid position. If an outer approximating polytope is not symmetric, replacing it by $P \cap (-P)$ preserves $C \subseteq P \cap (-P) \subseteq P \subseteq (1 + \eta)C$ and increases the number of facets by at most a factor of two.*

The next lemma is the elementary step that converts a symmetric polytope with M facet pairs into an intersection of M genuine centered ellipsoids, with arbitrarily small multiplicative loss. Geometrically, each slab is a degenerate centered ellipsoid, and we make it positive definite by adding a small multiple of $\|x\|_2^2$.

Lemma 3.3 (Replacing slabs by positive definite ellipsoids). *Let*

$$P = \{x \in \mathbb{R}^n : |a_j^T x| \leq 1, j = 1, \dots, M\}$$

be a bounded symmetric polytope. For every $\beta > 0$ there are positive definite matrices $Q_1, \dots, Q_M$ such that, for

$$K := \bigcap_{j=1}^M \{x \in \mathbb{R}^n : x^T Q_j x \leq 1\},$$

one has

$$K \subseteq P \subseteq (1 + \beta)K.$$

Proof. Let

$$R := \max_{x \in P} \|x\|_2 < \infty.$$

Since P is full-dimensional and bounded, $R > 0$. Choose

$$\lambda := \frac{(1 + \beta)^2 - 1}{R^2} > 0$$

and set

$$Q_j := a_j a_j^T + \lambda I_n, \quad j = 1, \dots, M.$$

Then $Q_j \succ 0$. If $x \in K$, then for each j ,

$$(a_j^T x)^2 \leq x^T Q_j x \leq 1,$$

so $x \in P$. Hence $K \subseteq P$.

Conversely, if $x \in P$, then $\|x\|_2 \leq R$ and, for every j ,

$$x^T Q_j x = (a_j^T x)^2 + \lambda \|x\|_2^2 \leq 1 + \lambda R^2 = (1 + \beta)^2.$$

Therefore $(x/(1 + \beta))^T Q_j (x/(1 + \beta)) \leq 1$ for every j , so $x/(1 + \beta) \in K$. Equivalently, $x \in (1 + \beta)K$. This proves $P \subseteq (1 + \beta)K$. $\square$

Proposition 3.4 (Upper bound). *For every fixed $n \geq 2$ there is a constant $C_n < \infty$ such that, for all sufficiently small $\varepsilon > 0$,*

$$m_n(\varepsilon) \leq C_n \varepsilon^{-(n-1)/2}.$$

Proof. Fix a symmetric convex body $C \subset \mathbb{R}^n$ and $0 < \varepsilon < 1$. Put

$$\eta = \beta = \varepsilon/3.$$

Then $(1 + \eta)(1 + \beta) \leq 1 + \varepsilon$ for $0 < \varepsilon \leq 1$. By Lemma 3.1, choose a symmetric polytope P with at most $A_n \eta^{-(n-1)/2}$ pairs of facets such that

$$C \subseteq P \subseteq (1 + \eta)C.$$

Set

$$P_0 := (1 + \eta)^{-1}P.$$

Then

$$P_0 \subseteq C \subseteq (1 + \eta)P_0.$$

Apply Lemma 3.3 to P_0 with parameter β . This gives an intersection K of the same number of positive definite centered ellipsoids such that

$$K \subseteq P_0 \subseteq (1 + \beta)K.$$

Consequently,

$$K \subseteq C \subseteq (1 + \eta)P_0 \subseteq (1 + \eta)(1 + \beta)K \subseteq (1 + \varepsilon)K.$$

The number of ellipsoids is at most

$$A_n(\varepsilon/3)^{-(n-1)/2} = A_n 3^{(n-1)/2} \varepsilon^{-(n-1)/2}.$$

Thus the desired upper bound holds, for instance with $C_n = A_n 3^{(n-1)/2}$. □

4 A connectedness lemma for quadratic superlevel sets

The lower bound is naturally stated in real projective space because all bodies and all quadratic forms involved are even. Let

$$\mathbb{P}^{n-1} := \mathbb{S}^{n-1} / \{u \sim -u\}$$

with projective angular distance

$$d_{\mathbb{P}}([u], [v]) = \arccos |\langle u, v \rangle|.$$

Lemma 4.1 (Projective connectedness of quadratic superlevel sets). *Let $A = A^T \in \mathbb{R}^{n \times n}$ and $t \in \mathbb{R}$. Then*

$$U_t(A) := \{[u] \in \mathbb{P}^{n-1} : u^T A u > t\},$$

where u is any unit representative of $[u]$, is either empty or connected.

Proof. The condition $u^T A u > t$ on unit vectors is equivalent to

$$u^T (A - tI) u > 0.$$

Replacing A by $A - tI$, it suffices to prove that

$$U := \{[u] \in \mathbb{P}^{n-1} : u^T A u > 0\}$$

is empty or connected. Let

$$\mathbb{R}^n = V_+ \oplus V_0 \oplus V_-$$

be the orthogonal decomposition into the positive, zero, and negative spectral subspaces of A . If $V_+ = \{0\}$, then $U = \emptyset$.

Assume $V_+ \neq \{0\}$. Every representative u of a point in U decomposes uniquely as $u = u_+ + u_0 + u_-$ with $u_+ \in V_+$, $u_0 \in V_0$, and $u_- \in V_-$. Since $u^T A u > 0$, necessarily $u_+ \neq 0$. Define, for $0 \leq s \leq 1$,

$$H_s([u_+ + u_0 + u_-]) := [u_+ + (1-s)(u_0 + u_-)].$$

This is well-defined in projective space. Moreover,

$$(u_+ + (1-s)(u_0 + u_-))^T A (u_+ + (1-s)(u_0 + u_-)) = u_+^T A u_+ + (1-s)^2 u_-^T A u_-.$$

The negative contribution is multiplied by a factor at most 1, while the positive contribution is unchanged. Hence the displayed quantity is at least $u^T A u > 0$. Thus H_s stays inside U and deforms U onto $\mathbb{P}(V_+)$. Since $\mathbb{P}(V_+)$ is connected, including the one-dimensional case where it is a single point, U is connected. $\square$

5 The lower bound

We now construct a symmetric convex body with many shallow, separated antipodal cap cuts. Each cut will force a distinct ellipsoid.

Let $0 < \varepsilon < 1/10$ and set

$$\delta := 2\varepsilon, \quad \alpha := \arccos(1 - \delta).$$

For small ε , $\alpha \asymp \sqrt{\varepsilon}$.

We use the following elementary packing fact.

Lemma 5.1 (Packing in projective space). *For each fixed $n \geq 2$ there are constants $b_n > 0$ and $\alpha_n > 0$ such that, for every $0 < \alpha < \alpha_n$, there exist points $[u_1], \dots, [u_N] \in \mathbb{P}^{n-1}$ with pairwise distances at least 2α and*

$$N \geq b_n \alpha^{-(n-1)}.$$

Proof. Let $M = \mathbb{P}^{n-1}$ with its standard Riemannian volume. For sufficiently small $r > 0$, every metric ball of radius r in M has volume at most $B_n r^{n-1}$, where B_n is a constant depending only on n . Choose a maximal 2α -separated subset $\{[u_1], \dots, [u_N]\}$ of M . By maximality, the balls of radius 2α centered at these points cover M . Therefore

$$\text{vol}(M) \leq \sum_{j=1}^N \text{vol } B_{\mathbb{P}}([u_j], 2\alpha) \leq N B_n (2\alpha)^{n-1}.$$

Thus

$$N \geq \frac{\text{vol}(M)}{B_n 2^{n-1}} \alpha^{-(n-1)}.$$

This proves the claim after setting $b_n = \text{vol}(M)/(B_n 2^{n-1})$. $\square$

Choose $[u_1], \dots, [u_N]$ as in Lemma 5.1. Define

$$C_\varepsilon := B_2^n \cap \bigcap_{j=1}^N \{x \in \mathbb{R}^n : |\langle u_j, x \rangle| \leq 1 - \delta\}. \quad (1)$$

This is a symmetric convex body. It contains $(1 - \delta)B_2^n$, so it is full-dimensional.

For $v \in \mathbb{S}^{n-1}$ its radial function is

$$r_{C_\varepsilon}(v) = \min \left\{ 1, \min_{1 \leq j \leq N} \frac{1 - \delta}{|\langle u_j, v \rangle|} \right\}, \quad (2)$$

with the convention that the quotient is $+\infty$ when $\langle u_j, v \rangle = 0$. Since the projective distance between distinct $[u_j]$ and $[u_k]$ is at least 2α , and $\cos \alpha = 1 - \delta$, we have

$$r_{C_\varepsilon}(u_j) = 1 - \delta, \quad j = 1, \dots, N.$$

Indeed, the j th slab is active in direction u_j , while all other slabs allow radius at least 1 in that direction.

Let

$$\mathcal{A}_j := \{[v] \in \mathbb{P}^{n-1} : d_{\mathbb{P}}([v], [u_j]) < \alpha\}, \quad \mathcal{A} := \bigcup_{j=1}^N \mathcal{A}_j.$$

The sets $\mathcal{A}_1, \dots, \mathcal{A}_N$ are pairwise disjoint. If $[v] \notin \mathcal{A}$, then $|\langle u_j, v \rangle| \leq \cos \alpha = 1 - \delta$ for all j , and hence

$$r_{C_\varepsilon}(v) = 1. \quad (3)$$

Proposition 5.2 (Lower bound). *For every fixed $n \geq 2$ there are constants $c_n > 0$ and $\varepsilon_n > 0$ such that, for all $0 < \varepsilon < \varepsilon_n$,*

$$m_n(\varepsilon) \geq c_n \varepsilon^{-(n-1)/2}.$$

Proof. Let C_ε be the body in (1). Suppose

$$K = \bigcap_{i=1}^m \{x \in \mathbb{R}^n : x^T Q_i x \leq 1\}, \quad Q_i \succ 0,$$

satisfies

$$K \subseteq C_\varepsilon \subseteq (1 + \varepsilon)K.$$

For $u \in \mathbb{S}^{n-1}$ write

$$q_i(u) := u^T Q_i u.$$

Since

$$r_K(u) = \min_{1 \leq i \leq m} q_i(u)^{-1/2},$$

the inclusion $K \subseteq C_\varepsilon$ gives $r_K(u_j) \leq r_{C_\varepsilon}(u_j) = 1 - \delta$ for each j . Therefore, for every j , there exists an index $i(j)$ such that

$$q_{i(j)}(u_j) \geq (1 - \delta)^{-2}. \quad (4)$$

Because $\delta = 2\varepsilon$, for every $0 < \varepsilon < 1/2$,

$$(1 - \delta)^{-1} = (1 - 2\varepsilon)^{-1} > 1 + \varepsilon,$$

and hence

$$(1 - \delta)^{-2} > (1 + \varepsilon)^2.$$

Set

$$t := (1 + \varepsilon)^2.$$

On the other hand, if $[v] \notin \mathcal{A}$, then $r_{C_\varepsilon}(v) = 1$ by (3). The inclusion $C_\varepsilon \subseteq (1 + \varepsilon)K$ implies

$$r_K(v) \geq \frac{1}{1 + \varepsilon}.$$

Consequently, for every $i = 1, \dots, m$,

$$q_i(v) \leq (1 + \varepsilon)^2 = t. \tag{5}$$

Indeed, if some $q_i(v) > t$, then $q_i(v)^{-1/2} < (1 + \varepsilon)^{-1}$, contradicting the lower bound on $r_K(v) = \min_i q_i(v)^{-1/2}$.

For each i , define the projective superlevel set

$$U_i := \{[v] \in \mathbb{P}^{n-1} : q_i(v) > t\}.$$

By (5), each U_i is contained in $\mathcal{A}$. By Lemma 4.1, each nonempty U_i is connected. Since $\mathcal{A}$ is a disjoint union of the open sets $\mathcal{A}_1, \dots, \mathcal{A}_N$, a connected subset of $\mathcal{A}$ can meet at most one of these sets.

However, (4) and the strict inequality $(1 - \delta)^{-2} > t$ imply that

$$[u_j] \in U_{i(j)}, \quad j = 1, \dots, N.$$

Thus the same index i cannot be assigned to two different cap pairs. Therefore $m \geq N$. Lemma 5.1 and the relation $\alpha = \arccos(1 - 2\varepsilon) \asymp \sqrt{\varepsilon}$ give

$$m \geq N \geq c_n \varepsilon^{-(n-1)/2}$$

for all sufficiently small $\varepsilon > 0$. □

6 Proof of the main theorem and the error formulation

Proof of Theorem 2.1. The upper bound is Proposition 3.4, and the lower bound is Proposition 5.2. The case $n = 1$ is immediate because every symmetric convex body in $\mathbb{R}$ is an interval centered at the origin, hence is itself a centered ellipsoid in one dimension. □

Proof of Corollary 2.2. For fixed $n \geq 2$, Theorem 2.1 says that there are constants $a_n, b_n > 0$ such that, for all sufficiently small $\varepsilon > 0$,

$$a_n \varepsilon^{-(n-1)/2} \leq m_n(\varepsilon) \leq b_n \varepsilon^{-(n-1)/2}.$$

For the upper bound on $\Delta_n(m)$, choose

$$\varepsilon = \left(\frac{b_n}{m} \right)^{2/(n-1)}$$

with a harmless adjustment of constants to ensure that ε is in the small range. Then $m \geq m_n(\varepsilon)$, so every body is approximable within factor $1 + O_n(m^{-2/(n-1)})$. Thus

$$\Delta_n(m) \leq C'_n m^{-2/(n-1)}.$$

For the lower bound, choose

$$\varepsilon = \left(\frac{a_n}{2m} \right)^{2/(n-1)}.$$

Then $m_n(\varepsilon) > m$ for all sufficiently large m , so there exists a symmetric convex body that cannot be $(1 + \varepsilon)$ -approximated by an intersection of m centered ellipsoids. Hence

$$\Delta_n(m) \geq c'_n m^{-2/(n-1)}.$$

The two inequalities prove the corollary. □

7 Conclusion

The fixed-dimensional tradeoff is therefore

$$\begin{aligned} \text{number of centered ellipsoids} &\asymp_n \varepsilon^{-(n-1)/2}, \\ \text{best worst-case error with } m \text{ ellipsoids} &\asymp_n m^{-2/(n-1)}. \end{aligned}$$

This is the same exponent as for approximation by polytopes with facets. The lower bound shows that, in the worst case and for fixed dimension, the extra quadratic parameters in each ellipsoid do not allow one ellipsoid to cover multiple separated shallow antipodal cuts. The topological reason is that a strict superlevel set of a quadratic form is connected in real projective space.

The result should be viewed as a solution of the fixed-dimensional asymptotic problem and as partial progress toward the broader dimension-dependent problem. Determining sharp or even useful constants as functions of n , or finding the optimal two-parameter behavior in (n, ε) , remains outside the scope of the argument given here.

References

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